Compilation of Environmental, Threat, and Animal Data for Cetacean Population Health Analyses Platform (CETACEAN)

Christina M. Amend (christinaamend@tamu.edu)
Texas A&M University, Department of Geography, Gulf of America Coastal Ocean Observing System (GCOOS)

INTRODUCTION

The Deepwater Horizon Oil Spill caused lasting ecological damage across the Gulf of America (GOA), with marine mammal populations among the most severely affected. Effective recovery requires access to quality, integrated datasets. However, these data have historically been fragmented across disparate systems. To address this challenge, the Gulf of America Coastal Ocean Observing System (GCOOS), in collaboration with the Natural Resource Damage Assessment (NRDA) Trustee Implementation Group (TIG), developed the Compilation of Environmental, Threat, and Animal Data for Cetacean Population Health Analyses Platform (CETACEAN).

METHODS

This dashboard was developed as a component of a larger GOA spatial data portal designed to organize, visualize, and disseminate cetacean-related geospatial information. The data portal integrates sighting datasets, environmental rasters, suitability grids, and supporting layers within an ArcGIS Online environment. The CET data portal is supported by a distributed collaboration between ArcGIS Online and an ArcGIS Enterprise deployment. In this workflow, datasets designated for public access are referenced across environments, enabling Hub Sites to aggregate cloud-hosted layers, enterprise datasets, and authoritative content into a unified interface while preserving controlled data management within the organization's enterprise datastore.

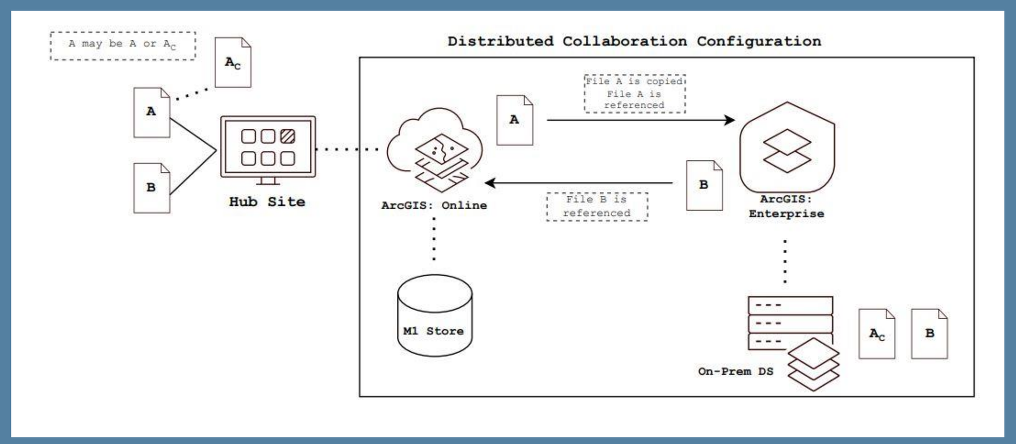

Figure I. Distributed collaboration workflow supporting the CET portal architecture.

RESULTS

Across the 12 modeled species, MaxEnt outputs demonstrated good discriminatory power, successfully identifying regions of high suitability consistent with known ecological distributions. Model response curves and variable contributions consistently highlighted depth, minimum sea surface temperature, and seafloor slope as dominant predictors. These results reflect the importance of bathymetric structure and oceanographic gradients in shaping cetacean habitat use throughout the GOA. The dashboard (Figure II) allows users to examine species-specific suitability patterns at both broad and fine spatial scales. Web maps were configured to host both the continuous raster surfaces and their associated point layers, ensuring consistent symbology and spatial alignment across species. Dashboard elements, including selectors, indicator panels, and summary charts, were linked through action-based filtering, allowing users to dynamically isolate species-specific outputs and explore spatial trends at multiple scales.

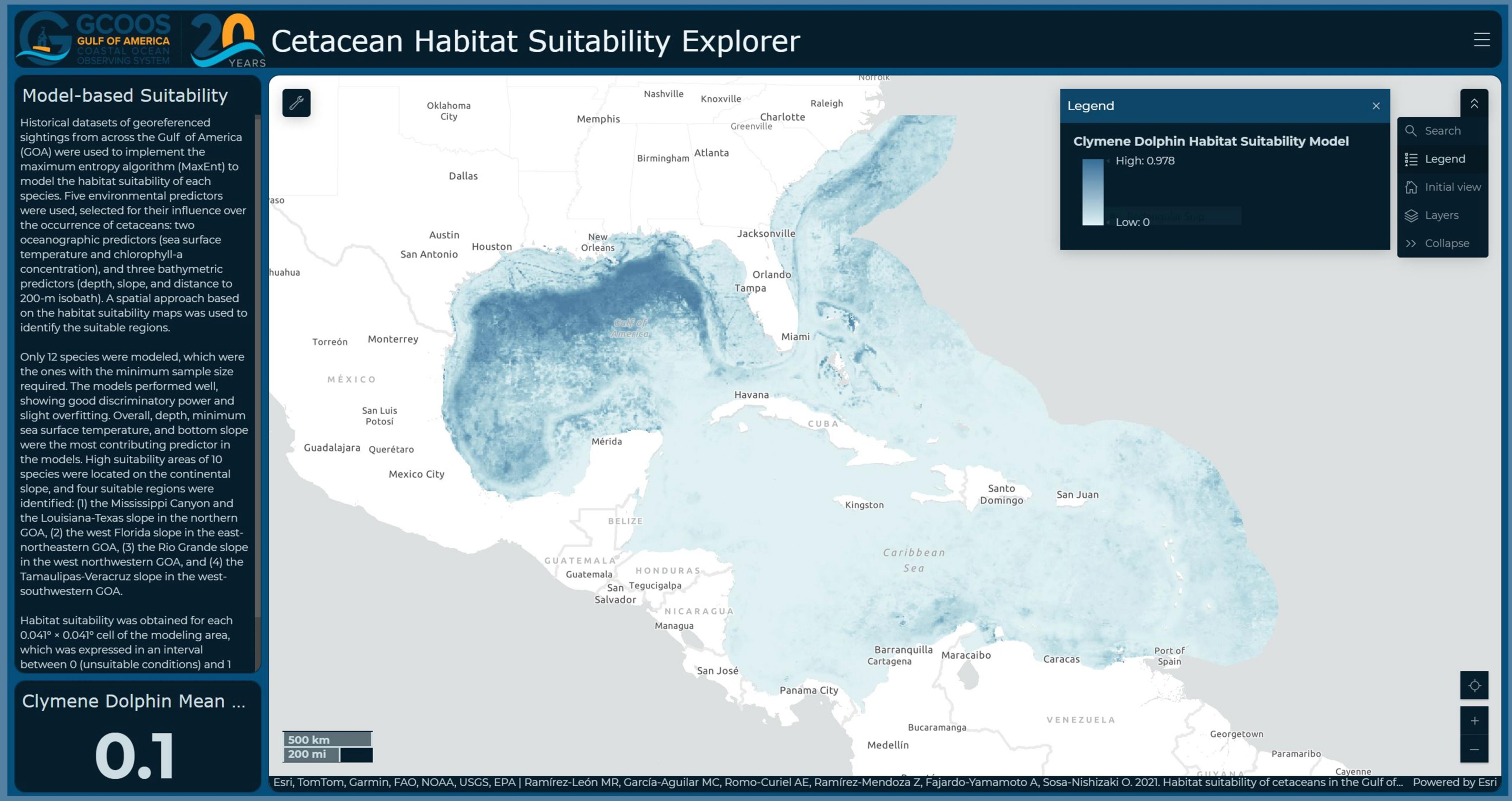

Figure II. ArcGIS Online dashboard used to visualize model outputs from MaxEnt-based cetacean habitat suitability analyses. Users can select species, examine continuous suitability surfaces, derive summary statistics, and identify regions of high predicted suitability.

The dashboard computes and displays cell-level and species-level metrics that summarize regional habitat quality. Key statistics include mean suitability, capturing overall habitat favorability across the species' modeled range. Users can click individual map cells to retrieve their precise suitability score, allowing localized assessment of habitat quality. These metrics collectively provide a quantitative understanding of habitat suitability distributions and highlight areas that may warrant conservation or further monitoring. The dashboard's interactive filters enable users to switch between species, compare suitability patterns, and explore differences in predictor-driven habitat use. The combination of imagery layers and point-based summaries provides a comprehensive depiction of cetacean habitat suitability. When integrated into the broader data portal, the dashboard supports spatial decision-making, ecological research, and regional marine management by presenting suitability information in an accessible, scalable, and visually interpretable format. This structure not only improves accessibility of the SDM results but also supports ecological research, management applications, and spatial planning by presenting complex model outputs in a clear, dynamic, and user-driven format.

DISCUSSION

The dashboard reveals clear spatial patterns of habitat suitability that align with major environmental and bathymetric features of the GOA. High suitability values for most species were concentrated along the continental slope, reflecting the influence of depth, minimum sea surface temperature, and seafloor slope. Four consistent hotspot regions emerged: the Mississippi Canyon and Louisiana–Texas slope, the west Florida slope, the Rio Grande slope, and the Tamaulipas–Veracruz slope. These areas correspond to zones of enhanced productivity and slope-associated ecological processes that support cetacean foraging. The dashboard's interactivity allows users to examine these patterns from regional scales to individual cells, improving interpretability compared to static maps. By summarizing distributional properties, it also supports rapid comparison of habitat quality among species. Integrated into the broader CET data portal, the dashboard helps synthesize model outputs with historical sightings and environmental layers, enhancing its value for research and spatial management.

Suitability Dashboard *CETACEAN Data Portal*

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